

When does cortical coherence carry shared content?

Compatibility and self-reconstruction as testable conditions

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Abstract

Synchrony accounts treat collective phase order as the mechanism that integrates distributed cortical activity. We show formally that phase order alone establishes neither shared content nor a self-model, and state what additionally does. Three conditions are separated: coherence (collective phase order), compatibility (agreement of independently decoded local descriptions where regions overlap) and self-reconstruction (accurate recovery of the global state from an internal encoding). Compatible local descriptions determine a unique global state. Coherence bounds disagreement between overlapping descriptions only through shared encoders of bounded sensitivity, and the bound grows as the square root of the population read: on a simulated cortical sheet it constrains at most 10 sites, chaining through overlapping patches extends it to 0.312 mm, and across distant regions the link from coherence to content is an assumption. A contracting self-map fixes one state but reconstructs accurately only states within a diameter set by its error and contraction factor. Lean 4 verifies

these implications; one model satisfies all connecting hypotheses at once, showing consistency, not cortical realization. The conditions yield a disconfirmable comparison: at matched phase coherence and single-region decoding accuracy, decoders fitted independently on two regions should agree more, beyond their separate accuracies, on perceived than on unperceived stimuli classified by a no-report marker. Choosing the overlapping regions by what each decodes makes the unity correlate operational relative to a declared decoder and criterion. An exploratory EEG analysis is a measurement check only. Unity and a minimal self remain interpretations that need an independent experiential endpoint.

1 Introduction

Endogenous extracellular fields can affect spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015) and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup’s location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver’s relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction from an internal encoding). The mathematical states are spatial measures and fields; identifying them with perceptual content requires independent

15 decoding evidence.

16 Binding-by-synchrony and communication-through-coherence accounts
17 treat phase relations as the medium of integration or of selective routing
18 (Singer & Gray, 1995; Fries, 2015), and field theories place integration in the
19 endogenous electromagnetic field (McFadden, 2020; Pockett, 2012). Both
20 leave open what fact about synchronized activity makes it one content rather
21 than coordinated activity carrying several. The contribution here is to sep-
22 arate the three requirements and prove which follow from which. Phase
23 order constrains shared content only through encoders related on overlaps,
24 with a bound that grows as the square root of the population it is read
25 on. Compatible local descriptions, not synchrony, determine a unique global
26 state. A contracting self-map fixes one state but can reconstruct accurately
27 only a family of bounded diameter. Each result names a measurement that
28 phase data alone cannot supply.

29 The objection that synchrony does not fix content is older than this work.
30 Critiques of the temporal binding hypothesis argued that synchrony, where
31 it occurs, cannot say which features belong together or what they jointly
32 represent: that grouping has to be computed before synchrony could sig-
33 nal it (Shadlen & Movshon, 1999). The formal results here are elementary:
34 uniqueness of gluing, Banach’s fixed-point theorem, the triangle inequality
35 and a pigeonhole count. What formalization adds to the informal argument is
36 quantitative and bookkeeping content. It gives the rate at which a coherence
37 bound on content weakens with population size and the population at which

38 it stops constraining anything, the diameter within which one contracting
39 map can reconstruct, and the number of codes a separated family of states
40 requires. It also gives a complete inventory of the premises a synchrony-to-
41 content argument consumes, each of which names a measurement, together
42 with one model in which all of them hold at once.

43 The familiar noisy mean-field oscillator supplies a stationary coherence
44 threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991). We add a resource
45 condition for a declared actuator, bounds that connect phase order to con-
46 tent only through specified encoders, and conditions under which compatible
47 descriptions glue and an accurate reconstruction fixes the glued state. A
48 Lean 4 development checks these implications and a joint witness satisfies
49 their eight connecting hypotheses in a specified model. It proves neither that
50 cortex satisfies those hypotheses nor that the glued state is experienced. The
51 latter identification motivates the framework and needs its own endpoint.

52 The empirical question is which measurements could support or discon-
53 firm the proposed connections. A phase resultant can miss a locally or-
54 dered winding field; stationary phase distributions can leave the coupling
55 mechanism unidentified; and coherent adaptation can lose alignment with
56 a designated content template. Independent decoders must identify shared
57 quantities before their agreement can be tested. We therefore distinguish sta-
58 tionary branches from finite observed onset, and compare prespecified field
59 and connectivity onset models on held-out emergence recordings. The com-
60 parison tests a specified spatial prediction, not the presence of experience.

61 Section 10.2 states the comparison that bears on conscious content directly.

62 Sections 2–5 develop the three requirements; Section 6 applies their mea-
63 surement obligations to a digital candidate. Section 7 states what the formal
64 composition establishes, and Sections 8–9 examine observables and the cor-
65 tical test. Table 1 lists the connecting hypotheses the article argues from;
66 Supplemental Material gives the full chain, derivations and controls. For a
67 theoretical article the Methods are the definitions, proofs and simulations.
68 Each result is stated in the section that uses it, Supplemental Material Ta-
69 ble S1 maps it to its Lean declaration, and every simulation’s model, param-
70 eters and controls are given in Supplemental Material. The Results are the
71 theorems of Sections 3–5 and the numerical controls of Section 8.

72 **2 Limits of phase order**

73 Several limits hold without the cortical hypothesis. Equal phases can coexist
74 with incompatible local descriptions; phase order constrains shared content
75 only through encoders related on overlaps. A winding field can have zero
76 global resultant while remaining locally ordered. A constant readout can have
77 a unique fixed point without reconstructing its input. A region encodes only
78 the states it distinguishes, and by a declared deadline a site can report only
79 changes inside its causal past. These limits reflect familiar phase oscillator
80 and information constraints (Acebrón et al., 2005; Ott & Antonsen, 2008;
81 Dörfler & Bullo, 2014; Breakspear, 2017). The sections below state each one

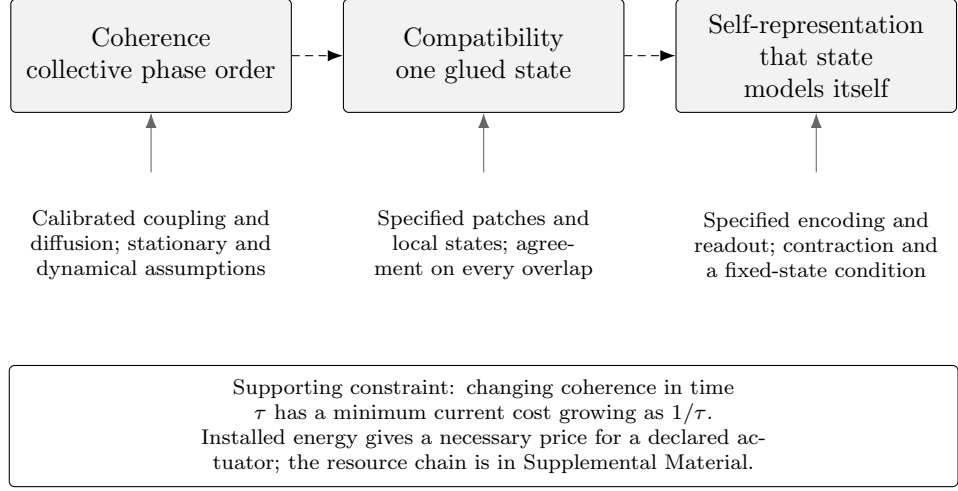


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the connecting hypotheses of the formal composition that the article argues from.

82 beside the condition or measurement it motivates.

83 3 Coherence: physical coupling and its cost

84 3.1 Dynamics and the stationary threshold

85 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

86 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
87 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
88 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
89 quency heterogeneity and diffusion are distinct parameters. The stationary
90 calculation below uses identical frequencies and this scalar coupling. Ap-
91 plying it to a spatial kernel requires an aggregation rule and a check of the
92 resulting approximation.

93 For identical frequencies in their rotating frame, the positive normalized
94 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

95 where I_n are modified Bessel functions and ψ is the collective phase. The for-
96 mal derivation uses the absence of stationary current for a periodic gradient
97 drift to obtain this density, then verifies normalization and self-consistency.
98 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz &
99 Mirollo, 1991). Above threshold, exactly one positive solution coexists with
100 zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

101 Figure 2 shows this stationary branch.

102 Linearization about the uniform density gives rate $K/2 - D$ for the first
103 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists

104 exactly above threshold. About the coherent density, the quadratic form
 105 is strictly dissipative transverse to collective rotation, whose rate is zero.
 106 These linear estimates concern the limiting equation. Relating them to finite
 107 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 108 tion; exchange symmetry alone does not establish the needed independence
 109 (Sznitman, 1991).

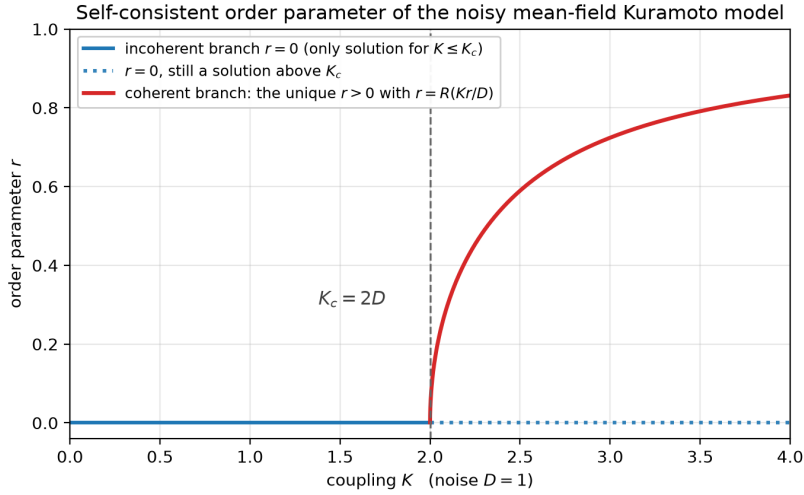


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

110 3.2 A speed limit on changes of coherence

111 One thermodynamic statement bears on coherence directly. For a smooth
 112 positive periodic phase density with probability current J and diffusion D ,

113 the current functional $\sigma_J = \int J^2/(D\rho) d\theta$ satisfies

$$\int_0^\tau \sigma_J dt \geq \frac{[\arcsin r(\tau) - \arcsin r(0)]^2}{D\tau} \quad (4)$$

114 while $0 < r < 1$. A change of coherence completed in time τ therefore has a
115 minimum current cost that grows as $1/\tau$, whereas holding a stationary order
116 has no positive floor. Read the other way, a current budget C sets a minimum
117 duration $\tau \geq [\Delta \arcsin r]^2/(DC)$ for a given change. This bears on ignition-
118 like transitions, in which large-scale coherence rises abruptly (Dehaene et al.,
119 2014), and on recovery latency after anaesthesia: each has a floor set by the
120 current available to drive it. The bound is a heat or power statement only
121 when J is identified with a physical flux, a reservoir at a stated temperature
122 is specified and local detailed balance holds for the steps concerned; only
123 then does σ_J convert to entropy production (Landauer, 1961; Norton, 2005;
124 Shenker, 2000). Installed energy supplies a separate, necessary price: for
125 a declared actuator whose response modes carry installation prices, mean
126 coupling is at most κU_{inst} for a mode-data factor κ , so a coherent stationary
127 branch requires installed energy $U_{\text{inst}} > 2D/\kappa$, and no cortical value of κ
128 is known. Supplemental Material gives both derivations, the register and
129 feedback budgets, and the sensitivity of the metabolic comparison to its
130 assumptions.

131 4 Compatibility: from coordinated activity 132 to a global description

133 4.1 Why common timing is insufficient

134 The formal setting is a sheaf: each patch of the substrate carries local de-
135 scriptions, and restriction maps cut a description on a patch down to any
136 smaller patch. A *section* is a description on one patch, and a *global section* is
137 a description on the whole substrate. For overlapping patches U and V , com-
138 patibility means that restricting either local description to $U \cap V$ gives the
139 same object. Descriptions can agree on shared quantities while contributing
140 different information elsewhere. Compatible sections of the specified sheaf
141 have a unique global extension. On a finite cover, local finite spatial mea-
142 sures agreeing as measures on every overlap also have a unique finite-measure
143 extension whose sheaf image is the glued section. No mass-one normalization
144 is imposed.

145 In the cup example, visual and reaching descriptions must express a
146 shared quantity, such as cup location, in common coordinates. Restriction
147 retains that quantity while dropping details particular to each description.
148 Agreement permits surface appearance and hand configuration to contribute
149 to one situation. The mathematical counterexample assigns the same phase
150 to every patch while two local mass profiles disagree on a shared site. Those
151 profiles have no common extension. Because they are chosen independently

152 of phase, synchrony places no constraint on their disagreement.

153 This distinction limits the biological interpretation. Receptive territories
154 and content domains must be related to spatial patches, and the descriptions
155 must support the required extension. Matching overlap marginals of de-
156 coded probability laws need not determine a unique global joint distribution
157 (Abramsky & Brandenburger, 2011).

158 The cover is itself a choice, and the construction constrains it in one way
159 only: synchronized patches agree where they overlap. Where they do not
160 overlap there is nothing to compare, so a cover by pairwise disjoint patches
161 satisfies that obligation for *every* assignment of local descriptions and phases
162 and the gluing theorem still returns a unique global section, which is the
163 tuple it was handed. Membership of the class then reports nothing about the
164 system beyond the patches covering it. What a gluing claim asks of a cover
165 is therefore a non-degeneracy the theorem does not assume: that distinct
166 patches meet, and that what they share is a quantity fixed independently of
167 the agreement it is being used to establish.

168 The cup example exposes a second gap, between two kinds of overlap. The
169 sheaf is built over a spatial substrate, so its patches overlap where they share
170 sites. Visual and motor territories that agree on cup location overlap in what
171 they describe, not in where they lie, and the construction supplies no relation
172 between the two. One route recasts the construction over content variables:
173 each region becomes a partial decoder whose domain is the set of variables
174 it estimates, and two regions overlap exactly where they estimate a common

175 variable. The gluing argument needs only a cover and restriction maps, so
 176 it applies formally to such a cover, but no result here is stated for one, and
 177 which variables and decoders form it is an empirical question. One principle
 178 selects such a cover without presupposing the agreement it is used to test.
 179 Assign each region the variables that can be decoded from its activity above
 180 a declared criterion, on data held out from the agreement test, and let two
 181 regions overlap on exactly the variables decodable from both. Decodability
 182 is fixed by what each region carries before any comparison is made, so the
 183 cover cannot be chosen to make agreement come out. It is non-degenerate
 184 exactly when its nerve is connected, meaning that every region is linked to
 185 every other through a chain of shared decodable variables, and recordings can
 186 fail that condition. The principle keeps free inputs: the decoder class, the
 187 criterion and the candidate variables, each of which must be declared before
 188 the agreement test. It therefore makes the unity correlate operational relative
 189 to those choices rather than absolutely, and carrying the formal results to
 190 such a cover is the recasting sketched above.

191 **4.2 When coherence does constrain content**

192 A declared relation between phase and content gives a positive result. Let
 193 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 194 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 195 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance,
 196 meaning that an encoder's output moves by at most L times the chord

197 between two input phases, and require $e_{i,x} = e_{j,x}$ wherever patches i and
 198 j overlap. Encoders may vary across sites; their equality on overlaps is a
 199 supplied relation between descriptions of the shared quantity. Write $z_k = e^{i\theta_k}$
 200 and $m = N^{-1} \sum_k z_k$, so $|m| = r$. The phasors' squared deviations from their
 201 mean sum to $\sum_k |z_k - m|^2 = N(1 - r^2)$, and $|z_i - z_j| \leq |z_i - m| + |z_j - m|$
 202 gives $|z_i - z_j|^2 \leq 2N(1 - r^2)$ for each pair. Consequently every overlap
 203 discrepancy is bounded by

$$\delta_r = L\sqrt{2N(1 - r^2)}. \quad (5)$$

204 Here discrepancy is the norm of the difference, and r and N are read on
 205 whichever population the bound is applied to. That choice is what decides
 206 whether the estimate says anything. A patch carries an order parameter
 207 of its own, and reading the bound there carries that patch's resultant and
 208 that patch's site count, so a large population observed through small patches
 209 is constrained by the small count. Read instead on the whole array, the
 210 factor $\sqrt{N}$ makes the estimate weak, because it draws a pointwise guarantee
 211 from an average over N sites. The order cannot be improved in general:
 212 one oscillator in antiphase among N leaves $N(1 - r^2)$ near 4. At $r = 1$ the
 213 observations agree exactly at either scale. A shared scalar encoder is a special
 214 case. Dropping the overlap relation permits disagreement even at perfect
 215 locking. A common nonexpansive linear projection preserves agreement, but
 216 agreement of a projection does not establish agreement of the full vector.

217 What Eq. (5) bounds is a chord, which never exceeds 2, so the estimate
 218 constrains anything only where the right-hand side falls below that. At the
 219 sheet's own patch order, the bound is informative for at most 10 sites. Three
 220 conditional results decline the step that puts $\sqrt{N}$ there. A pair count bounds
 221 the fraction of disagreeing pairs at any population size but names no pair.
 222 A chain across the sites that overlapping patches share compares distant
 223 territories through the patches between them, with error growing linearly in
 224 hops rather than with population: on the same sheet it stays informative
 225 for 19 hops, 0.312 mm. A spectral gap removes the population count at a
 226 frequency-locked configuration, given a declared algebraic connectivity of the
 227 coupling graph. Territories whose agreement unity concerns lie centimetres
 228 apart, beyond the chain's reach, and no measured connectivity yet supplies
 229 the spectral route, so across them the coherence-to-content link is the bridge
 230 assumption E78 (Table 1). Supplemental Material gives the three derivations
 231 and the sheet's numbers.

232 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 233 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 234 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 235 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (6)$$

236 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 237 duce an initial mismatch. Approximate observations leave a nondecaying

term in the error bound. Taking a uniform bound on Eq. (5) connects this update to the phase process.

4.3 What approximate agreement determines

Exact gluing requires exact restrictions. For nonnegative profiles on a finite substrate, suppose every overlap discrepancy is at most ε in uniform distance. Weights that vanish outside their patches and sum to one at each site select a global profile by weighted averaging. The selection lies within ε of each patch there, and any two such selections differ by at most ε , independently of cover multiplicity. An L -Lipschitz readout consequently varies by at most $L\varepsilon$ across selections.

At zero tolerance, the selection is the unique exact extension. At every positive tolerance there are families with no exact extension; selection then gives an approximation whose dependence on weights is bounded. Supplemental Material gives a three-site witness attaining the bound. The mass-profile construction uses the scalar case of the content dynamics and additionally requires nonnegativity of the initial profiles and observations. The general vector result supplies no identification with the spatial-measure sheaf.

These constructions concern declared objects: a specified cover, contents valued in a normed space, and encoders related on overlaps. They select neither the neural cover nor the decoder, and identifying their contents with perceptual content, or establishing that the identification bears on experience, is a further modelling task; Section 8 sets out what a measurement of

260 it would have to establish.

261 5 Self-representation: a state that models it- 262 self

263 5.1 The reflexive construction

264 Compatibility gives a global state. Reflexivity relates it to an encoding within
265 that state. Equip global sections with a complete metric. *Restriction reso-*
266 *nance* is the additional condition that the encoding is the state’s own restric-
267 tion to a region $\mathcal{A}$, so that the region encodes the state by being part of it.
268 Under that condition the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (7)$$

269 The region and readout are fixed across the state space. If F contracts and
270 fixes the section s_* formed from the local descriptions, Banach’s theorem
271 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
272 (Table 1 gives the full composition and its edges) supplies the contraction
273 and $F(s_*) = s_*$; restriction resonance is separate. Supercritical coupling
274 makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ less than one for
275 a relaxation time $\tau > 0$. The factor is stipulated, not derived. It borrows
276 the rate $(K - 2D)/2$ at which the first harmonic of the uniform density

grows (Section 3), a rate about a different state from the one F acts on,
and no result derives a Lipschitz law for an encoding or readout from the
phase dynamics. Constructing a map with that factor remains a modelling
obligation. A fixed contraction can reconstruct accurately only a restricted
family of states: if its error is at most ε throughout that family, any two
members are at most $2\varepsilon/(1 - \Lambda)$ apart.

In the cup example, the global state includes the represented scene and the
perceiver’s relation to it. A local encoding must retain enough information
for the readout to recover that modelled state; cup position alone would be
insufficient if hand configuration and viewpoint could not be reconstructed.

For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruction gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (8)$$

At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for
fixed positive ε and τ . Exact uniqueness survives while the error guarantee
weakens.

5.2 Reconstruction across a family of states

Sensitivity can be required quantitatively. Declare relevant macrostates, an
encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that

295 family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (9)$$

296 Thus M relevant states pairwise more than 2ε apart require at least M
 297 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 298 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 299 existing self-prediction map.

300 This is an additional accuracy criterion on a family, with no requirement
 301 for multiple exact fixed points. For the same fixed contracting map used
 302 by E89, however, the diameter bound above is an additional restriction: at
 303 zero error the family contains at most one metric-distinct state. A broad
 304 family therefore requires a different, input-conditioned reconstruction map,
 305 one map F_u with its own fixed point for each input u , with its own accuracy
 306 and stability assumptions. No such map is formalized or constructed here.
 307 Under E89 the proposed minimal-self correlate is therefore a condition on
 308 one self-state, not a model of a self that follows a changing scene. A blind
 309 encoder or constant readout fails the stated criterion for separated states
 310 despite having a unique fixed point.

311 Where the encoding is read as the state's own restriction to a region, the
 312 criterion becomes a condition on the region. Restriction resonance requires
 313 the encoder and that restriction to identify the same pairs of states, so a
 314 boundary reconstructing two relevant states more than 2ε apart while its

315 region reads them identically is not resonant. Read forwards, a resonant
 316 boundary that reconstructs a pairwise separated family restricts injectively
 317 on it. Which states a region reads identically is fixed by where the region
 318 sits, before an encoding is chosen, so the resolution the family demands is
 319 the region's and no readout supplies it. The implication runs one way: a
 320 region that does resolve the family is not thereby shown to carry a resonant
 321 encoding, and E89 supplies no such resolution.

322 Resonance and resolution are jointly satisfiable without a local fold. Take
 323 the region to be the whole substrate and use the identity readout: restric-
 324 tion is the identity, so this boundary is resonant and reconstructs every de-
 325 clared family exactly. Its map has Lipschitz factor one and does not satisfy
 326 E89's strict contraction premise on a nontrivial metric space. A properness
 327 condition on the region is therefore a separate requirement, and the fixed-
 328 contraction diameter bound remains in force when E89 is also required. With
 329 an L -Lipschitz readout, a family reconstructed within ε whose members lie
 330 pairwise more than $d_{\min}$ apart has δ -separated codes whenever $\delta L \leq d_{\min} - 2\varepsilon$,
 331 so the family is no larger than the δ -packing number of what the region car-
 332 ries. That is the shape a continuous family of states needs, and at the whole
 333 substrate it says nothing, because the region's range is everything. The gain
 334 L is a declared input, derived here for no physical readout, and the bound
 335 weakens without limit as the gain grows.

336 5.3 Agreement after a change takes communication

337 Snapshot compatibility does not specify how quickly descriptions respond
338 when a shared quantity changes. In a synchronous network whose updates
339 read only the site’s own state and declared neighbours’ messages, two initial
340 configurations agreeing on a site’s causal past through T rounds produce the
341 same state there at round T . The deadline consequence is the one Section 2
342 states, and it follows from the reconstruction bound in Section 5.2: a report
343 from that site is a readout of a code that the two interventions leave identical.

344 A three-site delay line witnesses failure after one round and exact report-
345 ing after two; cutting the path prevents delivery at every round. A report
346 may still happen to be correct about one fixed world. The bound concerns
347 responsiveness across interventions.

348 Throughout this section *prediction* names a reconstruction map, and the
349 relevant family of states, the metric, the tolerance, the communication graph
350 and the deadline are all declared inputs, as are the represented variables and
351 the fidelity counted as accurate. Reading the map as temporal neural pre-
352 diction would require dynamics; reading the resulting code count as physical
353 memory capacity would require a hardware model, which a state-dependent
354 readout would have to extend.

355 6 What the conditions ask of a digital candi- 356 date

357 A claim that a digital system, such as a large language model (LLM) run-
358 ning on graphics processing units (GPUs), has unified content or represents
359 itself commits its maker to the measurements this account asks of cortex.
360 The maker must identify the shared quantities the system’s parts describe,
361 show that decoders fitted independently on those parts agree about them,
362 and show that a declared local code separates the relevant family of states
363 and survives intervention. The indicator-properties approach assesses arti-
364 ficial systems against computational properties drawn from theories of con-
365 sciousness (Butlin et al., 2023); the conditions here are fewer and each is
366 a measurement rather than an architectural property. A digital system is
367 also where they can be measured today, with full access to its state and
368 exact interventions on it: fitting decoders for one shared quantity on two
369 separate layers, heads or token positions, and testing their agreement and
370 whether reconstruction survives intervention, would carry out the proposed
371 measurement directly.

372 One constraint follows from the architecture alone. Within one forward
373 pass of a causally masked model, position t reads only its causal past, so an
374 intervention outside that past cannot be reconstructed at t by that step. This
375 is the deadline result of Section 5 applied to the mask, and it bounds which
376 variables a position can encode about the rest of the sequence within one

377 step. A second constraint needs a premise the architecture does not supply:
 378 a linear encoder and readout factoring through d coordinates pay reconstruc-
 379 tion error at least the discarded spectral tail of the coupling operator, but
 380 query/key dimension does not supply such a bottleneck for softmax atten-
 381 tion. These results rank no architecture family, imply no general inferiority of
 382 GPU hardware and settle no question of machine experience. Supplemental
 383 Material gives the bottleneck derivation, a synthetic two-view task with its
 384 controls, the information ceiling of token outputs and the hardware scope.

385 **7 The formal composition**

386 The Lean development composes the three requirements into one conditional
 387 theorem. Its passive and feedback branches each accept eight connecting hy-
 388 potheses, running from finite capacity through a register, dissipation and a
 389 continuum coupling to the field, coherence, the cover and the self-prediction
 390 fixed point. Each edge is supplied rather than derived from the one above it,
 391 and the inferential content of the composition is its last step, where contrac-
 392 tion and a fixed glued section give uniqueness. Table 1 lists the four edges the
 393 article argues from; Supplemental Material gives the full chain, its thermo-
 394 dynamic edges and the register ledger of the feedback branch. Both branches
 395 also take a cover carrying exact overlap compatibility and a reflexive bound-
 396 ary specifying encoding and readout, and conclude that the glued section is
 397 the unique fixed point of self-prediction. Composing the edges establishes

398 that the premises are consistent, named in one place and machine-checked
399 together; it does not establish that self-representation follows from coher-
400 ence. Joint witnesses satisfy every edge of each branch in a specified model,
401 and establish no common cortical mechanism. The development declares no
402 additional axioms beyond its libraries’ foundational axioms.

Table 1: The four connecting hypotheses the article argues from, shared by both branches of the formal composition. Supplemental Material gives the full eight-edge chain and, in Table S1, maps the node-level results.

Connection	Status	Required assumption
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.

Table 1 (continued)

Connection	Status	Required assumption
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.

403 8 What the observations establish

404 The composition holds under its premises. Whether observation can estab-
405 lish those premises is a separate question, and the three requirements it
406 connects concern different objects: phase dynamics, local descriptions and
407 internal encodings. Their observables must retain these distinctions. The

408 following controls show which mechanisms remain indistinguishable, which
 409 temporal approximations fail, which states one observable cannot separate,
 410 and what independent content measurements must establish. Each is a finite
 411 numerical regime on a declared model, with full parameters and controls in
 412 Supplemental Material, and none of them calibrates a cortical quantity.

413 **8.1 Stationary distributions leave the mechanism un-** 414 **determined**

415 Equation (2) gives a concentration–coherence relation with no fitted shape
 416 parameter. Estimate r from circular resultant length and a from density
 417 shape: inverting R from the measured r would make agreement automatic.
 418 Even independently estimated agreement identifies a statistical family shared
 419 by different mechanisms.

420 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
 421 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
 422 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
 423 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
 424 and diffusion D_0 have that same density. Their stationary distributions co-
 425 incide window by window despite different causes.

426 Resolved dynamics supply additional information. In the ideal model
 427 with continuous latent paths, quadratic variation accumulates at rate $2D$
 428 and conditional increment means constrain drift; a filtered observation of

429 the phase needs its own observation model before either is read off. Separat-
 430 ing endogenous coupling from a matched common drive additionally requires
 431 a measured drive or controlled perturbation; identifying electromagnetic cou-
 432 pling requires evidence specific to that interaction. Supplemental Material
 433 gives the equivalence construction and its numerical check. Phase statistics
 434 pooled over time and over bipolar pairs cannot supply these estimates: on
 435 public sedation EEG (Bajwa et al., 2025), the pooled pipeline places a per-
 436 fectly phase-locked signal at $r = 0.0001$, so a spatial test needs an estimand
 437 that removes the common rotation and a montage whose effect on spatial
 438 phase is modelled; Supplemental Material reports this estimator exercise and
 439 the dependence-aware design.

440 **8.2 Finite ramps change the observable onset**

441 The square-root law in Eq. (3) describes stationary solutions, and a chang-
 442 ing coupling needs time for the phase distribution to approach them. In a
 443 deterministic mean-field control crossing the threshold, none of the scanned
 444 rates meets the declared tracking tolerance at onset, so a slow control vari-
 445 able supplies no general guarantee of branch tracking. In a finite-population
 446 ramp study ($N = 2000$, $D = 1$, 32 replicas, 8 speeds), faster ramps de-
 447 lay escape. The fitted delay exponent 0.444 has a replica-bootstrap interval
 448 $[0.398, 0.496]$ that excludes the ideal $1/2$, a deterministic reference shows that
 449 the escape criterion changes the fitted exponent, and no common asymp-
 450 totic exponent is established. Delayed passage through a noisy bifurcation

(Berglund & Gentz, 2002) makes the apparent exponent depend on ramp speed and fitting window. Supplemental Material gives the sweep and its figure, the fluctuating-coupling and population-size controls, and the signed-network crossing bracket.

The ideal delayed-onset calculation motivates a square-root scaling hypothesis; its exponent is generic to suitable bifurcations and does not identify the coupling medium. A within-subject emergence design can vary the propofol taper and test how an operational onset changes with that manipulation, but interpreting the result as a coupling-rate law requires a measured or calibrated relation between taper and $\dot{K}$. The escape criterion must be set relative to each recording's pre-threshold fluctuation scale. Section 8.1 describes how D could be estimated from resolved phase traces.

8.3 A winding state leaves the order parameter ambiguous

The global order parameter r discards spatial arrangement. On a periodic sheet, a phase field can wind once around the sheet and have $r = 0$ while neighbouring sites remain locked. At identical frequencies, every such winding is stationary for an isotropic symmetric kernel: contributions at opposite separations cancel. It violates the quarter-turn initial-spread premise of the convergence result, which is why that result can still select the uniform state.

In the spatial control, the winding persists below the interpolated de-

472 cay length 0.1871 mm and unwinds above it (Fig. 3). At 0.1 mm, uniform
 473 and winding starts under the same coupling, diffusion and kernel end at
 474 $r = 0.9657$ and $r = 0.0083$; patch-local order is 0.9015. Under quenched fre-
 475 quency spread 1.5 rad s^{-1} , the measured retention boundary is 0.1421 mm,
 476 and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen
 477 realizations. These are finite controls, not a cortical wave identification or a
 478 derived transition.

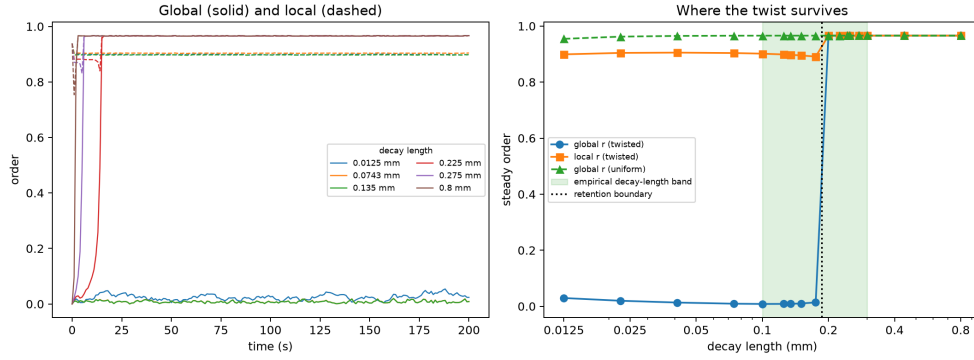


Figure 3: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

479 Linearization about a winding reweights each separation's coupling by
 480 the cosine of its phase advance. The resulting symmetric kernel has explicit
 481 mode rates: nonnegative reweighted coupling gives no growing mode, while

482 a positive rate destabilizes that winding. The criterion concerns linear rates,
 483 not asymptotic stability, and the measured retention boundary need not
 484 equal its limit. An amplitude extension adds an existence condition: $\Gamma + \lambda >$
 485 0, where Γ is linear gain and $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ comes from the same
 486 kernel. For the nonnegative kernels used here, a broad kernel can remove a
 487 winding from that model even though the phase-only equations retain it as
 488 a stationary state.

489 Neither model turns these states into measured cortical spirals. The pe-
 490 riodic sheet cannot carry a lone phase defect, the relaxational dynamics has
 491 no propagation speed, and the amplitude variable has no calibrated relation
 492 to a recorded envelope. Supplemental Material gives the spectral, amplitude
 493 and defect results. Locally, Eq. (5) can still bound content disagreement on a
 494 winding patch when shared encoders are supplied, although its global $r = 0$
 495 form is uninformative. A nonzero ring winding number q also obstructs a
 496 global real-valued phase lift; it has no identified content meaning here. Thus
 497 an onset criterion based only on r can register a change of basin as well
 498 as a change of coupling. The protocol of Section 9.3 needs a spatial phase
 499 observable.

500 8.4 Coherent adaptation can lose environmental align- 501 ment

502 An adaptive coupling rule might supply representational structure, but its
503 objective must reward the relevant agreement. For symmetric coupling, pair-
504 wise sine terms cancel in the population mean drift. Minimizing squared drift
505 at that fixed mean therefore rewards equalizing drift across nodes. In the
506 frequency-cluster control, this favors coupling between unlike-frequency clus-
507 ters, whereas the diagnostic environmental template rewards coupling within
508 clusters.

509 The formalization proves descent of the squared-drift objective for a spec-
510 ified gradient flow at fixed phases. Its functional differs from thermodynamic
511 entropy production: at a coherent stationary density the probability current
512 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
513 rial gives the operator, its gradient and the conditions of the descent result.

514 The finite control co-evolves noisy phases and symmetric, nonnegative
515 couplings at fixed total resource. Its template is a diagnostic covariance
516 of latent signals, never a training input. Across three illustrative seeds,
517 second-half order remains between 0.9167 and 0.9222, while the within-
518 cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–
519 0.6135 (Fig. 4). Random-update, permuted-gradient and frozen-kernel con-
520 trols are reported in Supplemental Material.

521 The immediate post-update objective reduction disappears when aver-

aged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

8.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on synthetic data separates these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed con-

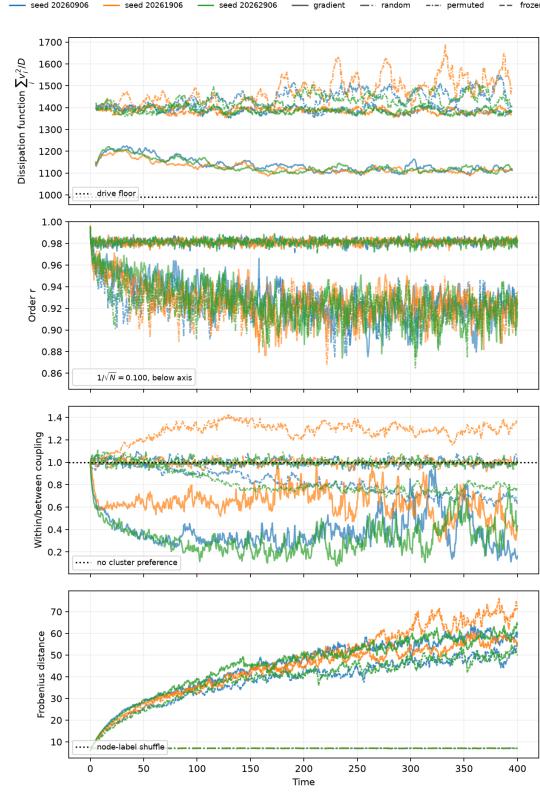


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

544 tent, phase carries a share 1.000 of the content entropy when content is built
545 from it and 0.000 when it is not. A phase-derived statistic reports com-
546 patibility on phase-locked patches whose local descriptions disagree. The
547 implemented content statistic detects their incompatibility. These controls
548 validate a measurement on known constructions; a neural content model and
549 decoder remain to be established.

550 Choosing the cover also chooses a candidate subject: trivial descriptions
551 can agree while carrying little content, and a partition can conceal disagree-
552 ment. The useful choice must therefore be constrained by what neural ac-
553 tivity encodes. The exact composition assumes compatibility of its supplied
554 cover. The content mechanisms above identify additional hypotheses under
555 which agreement can arise; they leave the empirical choice of content and
556 the connection to that cover explicit.

557 **9 The cortical hypothesis and an awakening** 558 **test**

559 The cortical proposal identifies a physical source of coupling and asks whether
560 its changes help explain recovery. The tests in Section 8 determine what
561 observations must distinguish before a phase trace can support this identifi-
562 cation.

563 9.1 Physical identification and calibration

564 The cortical hypothesis identifies part of effective coupling with endogenous
565 extracellular electromagnetic interactions. Fields can modulate spike timing
566 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
567 propose coordination of distributed neural information (Pinotsis & Miller,
568 2022; McFadden, 2020). Phase organization under propofol motivates study-
569 ing this regime (Bastos et al., 2021), but does not measure coupling.

570 The formal field predicate requires a jointly continuous kernel, a normal-
571 ized substrate, coupling equal to the kernel average, and no single site pro-
572 viding more than half that strength. Toy constructions satisfy it. A cortical
573 realization must specify the scale at which the kernel and its reduction ap-
574 ply: calcium imaging reports different state-dependent modularity patterns
575 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

576 Cortical columns are the obvious candidate unit, and they answer one
577 of the two demands the formalism makes. As the oscillator index they are
578 a serious proposal: they are delineated by response properties and anatomy
579 rather than by the agreement they would be used to establish (Mountcastle,
580 1997), and a phase can be estimated from a column-sized territory at the
581 electrode spacings used for cortical wave measurements (Townsend et al.,
582 2015). As the content cover they are a poor one, for three separate reasons.
583 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
584 the compatibility obligation without constraining anything (Section 4). The
585 cover condition additionally requires the patches to exhaust the substrate,

586 while much of association cortex has no accepted columnar parcellation. And
587 the column's standing as a canonical cortical unit is contested (Horton &
588 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
589 supplies the oscillator index; the content cover remains an open empirical
590 question, which the bridge assumption E78 records.

591 The spatial reduction is also testable. A row-sum aggregation at zero
592 diagonal tracks the uniform-coupling control over the tested spatial decay
593 range, while frequency heterogeneity displaces the operational threshold. The
594 fit window biases both estimates relative to $2D$; agreement with the control
595 is therefore the supported conclusion. Supplemental Material reports the
596 reduction, its numerical error and the distinct assumptions needed to identify
597 the kernel average with the scalar coupling of the installed-energy condition.

598 The field commitment requires physical evidence beyond a choice of math-
599 ematical representation, and the available support comparisons do not supply
600 it. A finite matrix spread over positive-mass cells preserves its total cou-
601 pling and phase correlation exactly, while point-supported vanishing under
602 an atomless measure and a disadvantage from excluding a better-correlated
603 pair concern specified support constraints; Supplemental Material reads the
604 same comparisons against a digital candidate. Neither supplies energy prices.

605 A spike-time shift alone supplies no coupling rate. The dimensionless
606 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and

607 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (10)$$

608 This rate describes how induced timing shifts accumulate into drift. Nei-
609 ther it nor the installation factor κ follows from a heat bound. Testing the
610 physical threshold requires independent estimates of the interaction kernel,
611 its aggregation and diffusion in the same cortical regime.

612 **9.2 Extracellular geometry and recovery**

613 The proposed control variable is extracellular geometry. Extracellular vol-
614 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
615 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
616 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
617 at $0.3\text{--}0.6 \text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
618 an effective-medium description predicts that a larger extracellular fraction
619 raises conductivity and reduces field amplitude.

620 The coupling-gated hypothesis consequently predicts that extracellular
621 contraction on awakening raises E and, holding the other factors in Eq. (10)
622 fixed, raises effective coupling. This argument supplies a direction; the
623 geometry–coupling relation, its magnitude and its role in human recovery
624 require measurement.

625 For a smooth crossing at nonzero speed, evaluating the stationary branch

626 along the changing control gives a deterministic tracking approximation that
 627 fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (11)$$

628 This is a deterministic tracking approximation from Eq. (3) at fixed D ; the
 629 crossing controls in Section 8.2 show that it fails as an observed-onset ap-
 630 proximation under the tested initialization. A separation of geometry and
 631 phase time scales can support tracking away from threshold. The awakening
 632 hypothesis needs a calibrated dynamical prediction for an operational onset
 633 above the finite-size fluctuation floor. Crossing K_c alone does not supply a
 634 recovery latency.

635 **9.3 A protocol for the awakening experiment**

636 Neural inertia describes a behavioural barrier and hysteresis between anaes-
 637 thetic induction and emergence (Friedman et al., 2010); recovery can pass
 638 through metastable activity states (Hudson et al., 2014; Proekt & Hudson,
 639 2018). The $v^{1/2}$ exponent of an ideal delayed bifurcation is generic (Baer
 640 et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field
 641 mechanism from a synaptic one. A possible discriminator is spatial onset
 642 relative to independently estimated local field coupling and network connec-
 643 tivity. The present mean-field model has no spatially varying κ and derives
 644 no nucleation order, so a field-based onset map needs a separate model spec-

645 ified before the observations are inspected.

646 For that comparison, partition the recording into prespecified regions i
647 and measure pre-onset order $r_i(t_0)$ and diffusion D_i . Define a field score
648 $h_i = N_i \mathcal{T}_i E_i f_i$ from independent pre-onset field and timing measurements,
649 and a connectivity score c_i from a separately estimated structural network.
650 For the same measured emergence drive $g(t)$, compare $K_i^F(t) = a_F g(t) h_i$ with
651 $K_i^C(t) = a_C g(t) c_i$. Each model has one fitted scale. The local first-harmonic
652 approximation predicts

$$\hat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (12)$$

653 Predicted onset is the first crossing of a region-specific threshold fixed from
654 pre-onset fluctuations. Fit the two scales on training sessions, then com-
655 pare held-out onset times and site order under the same observation model.
656 Larger field scores predict earlier onset only when baseline order and diffu-
657 sion do not reverse the ranking. This declared approximation omits interre-
658 gional coupling and saturation; a successful ranking would support its field
659 covariate, not prove ephaptic causation. Supplemental Material specifies the
660 controls and failure conditions.

661 A within-subject protocol can manipulate emergence rate, with spatially
662 resolved recordings to compare the proposed field and connectivity models:

- 663 **1. Manipulate emergence rate.** Vary the propofol infusion taper
664 across sessions within the same subject, recording drug concentrations

665 or a validated pharmacokinetic proxy to quantify the manipulation.

666 **2. Measure behavioural recovery.** Record time-to-response to a stan-
667 dard stimulus as the independent behavioural endpoint, rather than
668 estimating ΔK directly.

669 **3. Resolve spatial onset.** Identify anatomical locations where coher-
670 ence first crosses the baseline-scaled threshold and score the two maps
671 of Eq. (12) on held-out sessions. The two covariate maps are fixed
672 before the outcome sessions are inspected.

673 **4. Match physical controls.** Confirm that the observed coherence-to-
674 amplitude relation holds under the specific local diffusion D and ob-
675 servation noise. The model’s quadratic-variation rate $2D$ applies to
676 the latent phase; a band-limited analytic phase is smooth below its
677 correlation time, and on a constructed field its squared-increment rate
678 changes 37-fold between 0.2-ms and 20-ms steps. Estimating D there-
679 fore requires an explicit observation model for filtering and sampling.

680 Predeclare the recording window, regional fluctuation thresholds, and a
681 shared timing-error model. Fit the two scales and that observation model
682 on training sessions only; on held-out sessions, score the observed site-onset
683 times and sites that never cross by the same regional predictive rule. Report
684 site order as a secondary outcome and behavioural time-to-response sepa-
685 rately. Sessions without a crossing are censored observations, not zero delays
686 or exclusions. Compare the two scores only where independently estimated

687 field and connectivity maps predict distinguishable onset patterns at the
688 recording resolution. A held-out field-model advantage would support that
689 specified spatial predictor; it would not identify an ephaptic cause, shared
690 content or experience. Supplemental Material gives the controls and pilot
691 quantities needed to judge whether the comparison is informative.

692 **10 Discussion**

693 **10.1 Unity and a minimal self as interpretive motiva-** 694 **tion**

695 We propose the state glued from compatible local descriptions as a formal
696 correlate of conscious unity, and its accurate internal reconstruction as a cor-
697 relate of a minimal self. In the cup example, this would mean one description
698 that preserves the cup’s relation to the perceiver as well as the scene. Com-
699 patibility permits distinct contents and uncertainty; the shared quantities
700 must be chosen from independent evidence about what each region encodes.
701 The reconstruction need not be a verbal or autobiographical account.

702 The minimal self, in Gallagher’s sense, is the pre-reflective sense of be-
703 ing the subject of an experience, present before any narrative or reflective
704 self-description (Gallagher, 2000). Two of its features motivate the proposed
705 correlate. It is not a separate object of experience but a perspective built
706 into the experienced scene; correspondingly, the reconstruction is not a sec-

707 ond representation beside the glued state but a map whose fixed point is that
 708 state, read through a part of it. It is immediate rather than inferred from
 709 a higher-order state; correspondingly, the fixed point needs no further state
 710 that represents the first. The correspondence is structural and does not show
 711 that a fixed point is experienced as a perspective. Self-model theory locates
 712 the phenomenal self in a transparent model a system has of itself (Met-
 713 zinger, 2003), and interoceptive-inference accounts ground it in prediction of
 714 the body’s own states (Seth, 2013). The present condition is narrower than
 715 either: it asks only that a declared local encoding recover the global state,
 716 including perceiver-relative variables, at a declared fidelity, and it says noth-
 717 ing about transparency or about which bodily variables must be modelled.
 718 Either account could supply the family of states and the fidelity that the con-
 719 dition leaves as inputs. Under E89 that family is one state at zero error and
 720 a family of diameter at most $2\varepsilon/(1 - \Lambda)$ within error ε (Section 5.2), so the
 721 condition as proved constrains a single self-state; a self that persists through
 722 change needs the input-conditioned map that is not constructed here. Its
 723 proposed relation to experienced perspective remains an interpretation. Nei-
 724 ther identification is among the connecting hypotheses of the composition
 725 (Table 1), and satisfying them would not measure experience. The unity
 726 proposal is also only as definite as its cover. The gluing theorem holds for
 727 whatever cover it is given, a disjoint cover satisfies it trivially, and the cover
 728 of interest overlaps in content rather than in space. Section 4 proposes fixing
 729 that cover by decodability on held-out data, which excludes covers chosen

730 for agreement and makes non-degeneracy a testable property of the record-
731 ings. The decoder class and criterion remain declared inputs, and the formal
732 results are stated for spatial covers.

733 **10.2 A comparison that could disconfirm the link to** 734 **content**

735 The framework predicts that conscious content tracks compatibility, not co-
736 herence. The comparison must also separate compatibility from how well
737 each region encodes the stimulus, because two accurate decoders agree auto-
738 matically. If decoders d_A and d_B estimate a shared stimulus quantity x with
739 errors $e_A = d_A - x$ and $e_B = d_B - x$, then

$$|d_A - d_B| \leq |e_A| + |e_B|, \quad |d_A - d_B|^2 = |e_A|^2 + |e_B|^2 - 2\langle e_A, e_B \rangle. \quad (13)$$

740 Consciously perceived stimuli are usually decoded better, so higher agreement
741 on those trials would follow from single-region fidelity alone. At matched
742 single-region error, agreement differs only through the covariance of the two
743 errors, which is the term compatibility bears on: two regions describing one
744 content should err together.

745 Present matched stimuli that are perceived on some trials and not on
746 others, for example at masking or rivalry threshold, and fit decoders for one
747 shared stimulus quantity independently on two regions that both encode it,
748 selected by the decodability criterion of Section 4 on separate data. Clas-

749 sify trials by perceptual state with a no-report measure, such as reflexive
750 eye movements or pupil responses that track the dominant percept in rivalry
751 (Tsuchiya et al., 2015), so that the classification does not also select for the
752 neural correlates of reporting; report-based classification is a secondary anal-
753 ysis. Match the two classes on phase coherence between the regions and
754 on each region’s own decoding error (or, where the stimulus value is uncer-
755 tain, on each region’s signal strength). The statistic is the excess agreement
756 over a fidelity-matched null: pair region A ’s decode on one trial with region
757 B ’s decode on another trial of the same stimulus and error stratum, which
758 preserves both marginal accuracies and removes any coupling of the errors.

759 The proposed link predicts a larger excess on perceived trials at matched
760 coherence and matched single-region accuracy (Fig. 5). Equal excess across
761 classes, or excess that follows coherence and not perceptual state, counts
762 against it. A positive result is not yet specific: noise shared upstream of both
763 regions also correlates their errors, so the excess should be compared against
764 an upstream region’s decoding error, matched in the same way. A self-model
765 comparison has the same form: at matched accuracy for the scene variable,
766 the reconstruction of perceiver-relative variables, such as hand position rel-
767 ative to the target, from a declared local encoding should be more accurate
768 on perceived trials. These are correlational tests of proposed correlates, and
769 the no-report classification is itself an indirect marker of experience.

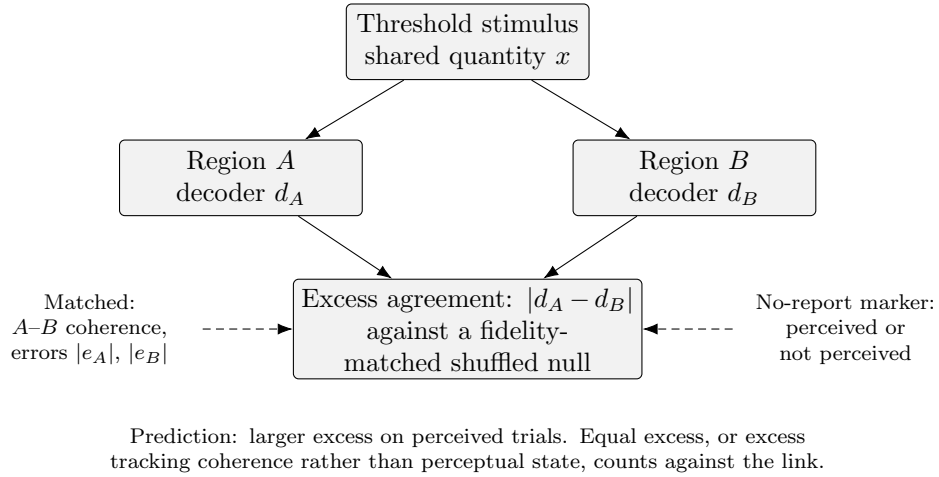


Figure 5: Design of the disconfirming comparison. Decoders for one shared stimulus quantity are fitted independently on two regions. Trials are classed by a no-report marker of perceptual state and matched on inter-regional phase coherence and on each region’s own decoding error, so that agreement cannot follow from single-region fidelity (Eq. (13)). The statistic is agreement in excess of a null that pairs decodes across trials within the same stimulus and error stratum.

10.3 What a test of the full proposal must establish

A test needs separate estimates of effective coupling and diffusion, shared content on overlapping regions, and reconstruction from a specified encoding at a declared fidelity. The cover, relevant state family and decoder must be fixed without selecting them for agreement. An independent experiential endpoint is needed to evaluate the interpretation; absence of a report alone does not define absence of experience. A unified episode with incompatible decoded descriptions would challenge the unity identification. A unified episode lacking the proposed self-encoding would challenge the minimal-self claim. Conversely, satisfying the mathematical conditions without such an

780 episode would challenge their sufficiency as correlates.

781 For the physical route, cortical actuator modes and prices, the geometry–
782 coupling relation and a dynamical onset model remain to be measured. The
783 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-
784 diction. Controlled changes in coordination and independently decoded con-
785 tent agreement would address the representational connection. The synthetic
786 decoder exercise calibrates part of these measurements, and the EEG exercise
787 identifies what a pooled bipolar estimand cannot measure; neither identifies
788 a neural content code or tests experience.

789 **10.4 Explanatory scope**

790 Integrated-information, global-workspace and predictive-processing accounts
791 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
792 1997; Dehaene et al., 2014; Friston, 2010). This account specifies conditional
793 relations among coupling, local content and reconstruction. Its formal field
794 condition requires a continuous kernel, a normalized substrate, a specified
795 average coupling and noise, and a site-contribution bound. Those conditions
796 do not identify an electromagnetic substrate; the extracellular interpretation
797 needs the separate calibration of Section 9. The joint witness shows that
798 the formal hypotheses can hold together in a declared model. A cortical
799 realization and an experiential interpretation require additional evidence.

800 Four limitations bound every claim above. Synchrony is not itself an
801 experiential measure, and nothing here establishes a consciousness criterion:

802 the conditions are proposed correlates to be tested against report or another
803 independent endpoint. The present EEG analysis checks a measurement
804 pipeline and does not validate any experiential claim. The formal results are
805 conditional on declared encoders, covers and maps whose cortical counter-
806 parts are unmeasured. The cover on which compatibility is assessed is fixed
807 only relative to a declared decoder class and criterion, and the formal results
808 are stated for spatial covers rather than for the content cover that criterion
809 yields.

810 The remaining explanatory question is whether these relations occur in
811 neural activity and matter for experience. The framework identifies candidate
812 correlates and tests of their physical and representational premises. It does
813 not explain why satisfying them would be accompanied by experience.

814 11 Conclusions

815 Phase coherence, compatibility and self-reconstruction are separable condi-
816 tions. Coherence constrains shared content only through encoders shared on
817 overlaps and with a bound that grows as the square root of the population
818 read. That bound is vacuous beyond 10 sites at the simulated sheet's patch
819 order; chaining it through overlapping patches reaches 0.312 mm of that
820 sheet, and a spectral route removes the population count only for a declared
821 connectivity at a frequency-locked configuration. Across the centimetre sep-
822 arations of distinct cortical territories, the positive link from coherence to

823 content is the bridge assumption E78 rather than a derived result. Com-
824 patible local descriptions, not synchrony, determine one global state; and a
825 contracting self-map fixes that state while reconstructing only a family of
826 bounded diameter. Each result replaces an assumption about synchrony by
827 a measurement: independently decoded agreement on overlaps, a declared
828 encoding and its fidelity, and calibrated coupling and diffusion. The compar-
829 ison of Section 10.2 states how such measurements could count against the
830 proposed link to conscious content.

831 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	Eq. (1), Supp.	Phase diffusion in inverse time; in the Supplemental Material's convergence proof, the phase spread $D \leq \pi/2$
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring

Table 2 (continued)

Symbol	Defined	Meaning
U_{inst}, κ	Section 3.2	Installed energy of a declared actuator and the mode-data factor bounding its mean coupling by κU_{inst}
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (9)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (7)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time; in Eq. (4), τ is the duration of a change
J, σ_J, C	Eq. (4)	Probability current of the phase density, its current functional and a current budget
T	Section 5	Number of communication rounds to a deadline
ν	Supplemental Material	Fan-in of a communication graph: the most sites any one site hears from
$\gamma, \mathcal{T}, f$	Eq. (10)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover's nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (9)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (7)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d

Table 2 (continued)

Symbol	Defined	Meaning
E	Eq. (10)	Field amplitude, in the cortical identification
s, s_*	§4–5	A global section and the one the reconstruction fixes

832 Data and code availability

833 The Lean development, simulation scripts, tests and saved numerical sum-
834 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
835 `cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and Math-
836 lib commit `5ce203ecd53f8c3e47ee5530d7c86b94a55f0183`. Publication
837 simulation values are generated from saved summaries without rerunning
838 production sweeps. The exploratory EEG analysis uses OpenNeuro dataset
839 `ds005620` (Bajwa et al., 2025) (subjects 1010, 1016, 1022, 1033, 1045,
840 1054, 1060, 1067, resting awake and sedation runs 1–2); preprocessing
841 and selection are described in Supplemental Material and implemented in
842 `simulations/empirical_collapse.py`. The repository contains the repro-
843 ducible analysis and does not redistribute the raw recordings.

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846 ing interests.

847 Use of AI tools

848 AI assistance to this work was substantial and is not separable by task.
849 Several coding assistants and chat interfaces were used in an interleaved way
850 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
851 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra and Sol; GPT-
852 5.6 Terra, Luna and Sol), Claude Code (Claude Opus 5 and Opus 5.5),
853 Gemini Pro through its web interface, and Anthropic’s Fable and DeepSeek
854 v4 Flash through OpenRouter, accessed with the Hermes CLI. Between them
855 they contributed to the Lean development, the simulation and analysis code,
856 and the drafting and revision of both documents. Because the tools were
857 used in combination and in alternation, no section, proof, figure or result is
858 attributable to any one of them, and none is claimed to be. The author is
859 responsible for all content, including everything a tool produced. Machine
860 checking establishes the stated Lean results under their hypotheses; it does
861 not validate biological interpretations or replace scientific review of generated
862 code and prose.

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